



inara
Technologies

Intelligent
Programmed
Networks

Software Requirements Specification Internship Management System

Version: 1.0

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1. Introduction

1.1 Purpose

The purpose of the Internship Management System (IMS) is to digitize and simplify the complete internship process at Inara Technologies. The system will manage internship applications, onboarding, training progress, weekly reports, project assignments, supervisor feedback, and historical internship records through a centralized web portal.

1.2 Scope

The system will provide different interfaces for Applicants, Interns, Supervisors, and Administrators.

It will allow applicants to apply for internships online, enable administrators to review and onboard selected candidates, assist interns in submitting required documents and weekly reports, and allow supervisors to monitor intern progress and provide feedback.

The system will also maintain historical internship records for future reference.

2. Objectives

- Eliminate manual internship management.
- Provide an online internship application system.
- Simplify intern onboarding.
- Track internship progress.
- Manage project assignments.
- Store internship history.
- Improve communication between interns and supervisors.

3. User Roles

3.1 Applicant

- Register/Login
- Apply for internship (Fill information, choose internship mode, department)
- Upload CV
- Track application status

3.2 Intern

- Login
- Upload required documents
- Download department training PDF
- Submit weekly reports
- View supervisor feedback
- View assigned projects
- Update assigned project progress
- Receive notifications

3.3 Supervisor

- View assigned interns
- Review weekly reports
- Approve or request changes
- Provide feedback
- Create and manage projects
- Assign interns to projects
- Track internship progress
- Upload department training PDFs
- View historical internship records

3.4 Admin

- Manage internship applications
- Approve or reject applicants

- Assign departments
- Manage users
- Verify submitted documents
- View historical internship records
- Send system notifications

4. Functional Requirements

Module 1 – Authentication & Role Management

- User Registration
- Secure Login
- Forgot Password
- Role-Based Access Control (RBAC) Roles:
 - Applicant
 - Intern
 - Supervisor
 - Admin

Module 2 – Internship Application

Applicants can:

- Complete application form
- Upload CV
- Select preferred department (optional)
- Select internship mode (Remote, Hybrid, On-site)
- Submit application
- Application Status:
 - Submitted
 - Under Review
 - Interview Scheduled
 - Selected
 - Rejected

Module 3 – Applicant Review

Admin can:

- View applications
- Search applicants
- Filter applications
- Change application status
- Send acceptance/rejection emails

Module 4 – Internee Onboarding

Selected applicants can be converted into internees.

HR/Admin can:

- Create internee account
- Activate portal access
- Initiate onboarding

Module 5 – Document Management

Intern uploads required documents:

- CV
- Academic Transcript
- Experience Certificates (Optional)
- CNIC/Identification (Optional)
- Document Status:
 - Pending
 - Uploaded
 - Verified
 - Rejected

Module 6 – Department Assignment

Admin assigns interns to departments such as:

- DevOps
- SysOps
- Security
- Networking
- Development

Interns can view their assigned department.

Module 7 – Training PDF Management

Admin uploads one official training roadmap PDF for each department.

Interns can:

- Download the training PDF
- View upload/update date

The system does not provide quizzes or learning content. Training is completed externally according to the roadmap provided.

Module 8 – Weekly Report Submission Interns

submit weekly reports including:

- Week Number
- Work Completed
- Challenges Faced

Attachment report Status:

- Submitted
- Under Review
- Approved
- Changes Requested

Module 9 – Feedback & Progress Tracking

Supervisor can:

- Review reports
- Provide written feedback
- Approve reports
- Request modifications

The system automatically displays internship progress.

Example:

- 5 / 12 Weeks Completed
- 42% Progress

Interns can also view previous supervisor's remarks.

Module 10 – Project Management

Supervisor can:

- Create projects
- Add project members
- Assign project tasks
- Monitor project progress

Interns can:

- Update task status
- View assigned projects
- Upload project deliverables

Module 11 – Historical Records

The system stores completed internship records including:

- Intern Information
- Internship Year
- Department
- Supervisor
- Weekly Reports
- Feedback
- Projects
- Completion Status
- CV

Admin can search records using:

- Name
- Department
- Year
- Supervisor

Module 12 – Notifications

System notifications are generated for:

- Application status updates

- Department assignment
- Weekly report feedback
- Project assignment
- Document verification
- Upcoming report deadlines

Notifications are displayed within the portal and may also be sent via email.

5. Non-Functional Requirements

Performance

- Page response time should be less than 3 seconds.
- Support multiple users simultaneously.

Security

- Password hashing
- Secure authentication
- Role-based authorization
- Protected file uploads

Reliability

- Regular database backups
- Error handling and validation

Usability

- Responsive interface
- User-friendly navigation
- Simple dashboard for each role

Maintainability

- Modular architecture
- Clean API structure
- Easy future enhancements

6. Business Rules

- Applicants do not require login.
- Only selected applicants become internees.

- HR/Admin controls application status.
- HR/Admin verifies all uploaded documents.
- Each internee belongs to one department at a time.
- Department Heads can only manage internees assigned to their department.
- Applications cannot be modified after submission unless reopened by HR/Admin.
- Rejected applicants may reapply after a defined cooldown period
- Intern portal access is deactivated upon internship completion
- Verified documents cannot be re-uploaded without HR/Admin reopening the request
- Department reassignment requires HR/Admin approval

7. Technology Stack

Frontend

- React.js
- HTML5
- CSS3
- Bootstrap 5 (or Tailwind CSS)

Backend

- Node.js
- Express.js

Database

- MariaDB

Authentication

- JSON Web Tokens (JWT)
- bcrypt.js for password hashing

File Storage

- Multer (file upload middleware)
- Local storage (with support for cloud storage in future)

Email Service

- Nodemailer (SMTP)

API

- RESTful API

Version Control

- Git
- GitHub

Development Tools

- Visual Studio Code

8. Future Enhancements

- Attendance Management
- Leave Request System
- Certificate Generation
- Dashboard Analytics
- Calendar Integration
- Cloud File Storage (AWS S3, Azure Blob Storage, etc.)
- Mobile Application

9. Conclusion

The Internship Management System will provide a centralized platform to manage the complete internship lifecycle - from application and onboarding to progress tracking, project management, and historical record maintenance. The system aims to improve efficiency, transparency, and communication while reducing manual administrative work and providing organized experience for applicants, interns, supervisors, and administrators.
